Utku Demirtaş

Software Engineer

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SUMMARY

Backend-focused software engineer building real-time engines, event-driven services and RAG pipelines, plus the CI/CD and containers that ship them. Computer engineering student working across Node/TypeScript, Spring Boot/Java and Python.

EXPERIENCE

Founder & Lead Developer *Okut Gitsin — okutgitsin.com (launching soon) | Nov 2025 – Present*

- Designed and built a live-stream vehicle auction platform end to end, from data model to deployment
- Engineered an atomic bid engine on Redis and Lua for millisecond-level, race-free bidding
- Built real-time bidding over WebSocket and live video auctions with LiveKit
- Added an escrow payment flow (held, refunded, forfeited) with IBAN verification, Helmet CSP/HSTS, rate limiting and GitHub Actions CI/CD

DevOps & Backend Developer *Extramus | Jul 2025 – Sep 2025*

- Built HR-management backend components and document-management infrastructure
- Implemented secure file-upload mechanisms with role-based access control (RBAC)
- Automated testing and deployment with GitHub Actions CI/CD pipelines
- Containerized backend services with Docker for consistent environments

R&D Engineering Intern *Telenity (R&D) | Jun 2024 – Aug 2024*

- Developed backend modules with Spring Boot
- Worked with Java, MongoDB, Kafka and SQL-based systems
- Focused on bug fixing, performance optimization and system stability

VOLUNTEERING EXPERIENCE

Educator & Co-leader *Anticverse Tech | 2022 – 2023*

- Helped manage the cybersecurity team and delivered training on data and web security

Team Leader & Design Engineer *Teknofest – Ouroboros Team | 2021 – 2022*

- Led the drone design and development process and managed technical coordination within the team

EDUCATION

B.Sc. Computer Engineering Konya Food and Agriculture University | 2022 – 2026

PROJECTS

Reveil — AI-Assisted Self-Management Platform *Ongoing*

- Authentication, behavior tracking and AI-assisted features via prompt-based workflows over a NestJS and FastAPI backend. — React Native, NestJS, FastAPI, PostgreSQL, OpenAI API

Costsight — Cloud-Cost Anomaly Detection

- Anomaly detection on AWS CUR data using STL, Isolation Forest and Z-Score detectors, with severity-banded alerts and a Streamlit dashboard. — Python, scikit-learn, Streamlit, AWS

Distributed Rate Limiter

- API gateway with token bucket, sliding window and fixed window limiters, backed by atomic Redis Lua scripts so limits hold across replicas. — TypeScript, Redis, Lua, Prometheus, Grafana

Event-Driven Order System

- Five microservices coordinating an order saga, with compensating transactions, idempotent consumers and one OpenTelemetry trace per order. — TypeScript, RabbitMQ, OpenTelemetry, Docker

Codebase RAG Q&A

- Repo answers with file:line citations: tree-sitter AST chunking, hybrid BM25 and vector retrieval, and an eval harness (94% hit@3). — Python, FastAPI, tree-sitter, pgvector

Open-Source Collaborations

- Contributed test suites and CI to Go, Rust, C++ and ML repositories, including k8s-resilience-harness, femheat, gochain and rust-url-shortener.

TECHNICAL SKILLS

Programming Languages: Java • Python • JavaScript • TypeScript • SQL • Verilog

Frontend: Next.js • React • Tailwind CSS

Backend: Node.js • Express.js • NestJS • Spring Boot • REST APIs

Security & Realtime: OWASP • Wireshark • Burp Suite • WebSocket • LiveKit

Databases: PostgreSQL • MongoDB • Redis • Prisma

DevOps & Tools: Docker • GitHub Actions • Kafka • RabbitMQ • Prometheus • OpenTelemetry • Linux • Postman

Cloud & AI: AWS • AWS ML Foundations • OpenAI API • FastAPI

Languages: English – C1 • Spanish – A1 • Turkish – Native